Summary Statistics and Descriptive Plots

Joël de Vries

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Data overview

The estimation sample contains **1030 rows** (participant × week combinations) for **17 participants**, after excluding three participants who did not meet the inclusion criteria (minimum 10 weeks of observations and minimum 500 videos per week on average; excluded: p05, p10, p19). The observation period runs from **25 november 2024** to **30 maart 2026**.

Table 1: Number of observation weeks per participant

Participant	Number of weeks
p01	65
p02	66
p03	66
p04	46
p06	66
p07	65
p08	11
p09	65
p11	58
p13	64
p14	66
p15	66
p16	65
p17	65
p18	66
p22	65
p23	65

Summary statistics

The four research variables summarized across all participants and weeks in the estimation sample.

Table 2: Descriptive statistics per research variable

Variable	N	Mean	SD	Min	Max	N missing
Discovery Ratio	1030	0.454	0.144	0.097	1.000	0
Avg. Session Duration (min.)	1030	21.295	10.238	1.133	93.077	0
Like-Rate	1029	0.026	0.053	0.000	0.600	1
Search Intensity	1030	24.592	26.579	0.000	172.000	0

Descriptive plots

Timeline per participant

Each line represents one participant. The timeline shows how the variables develop over the observation period. Notable trends and outliers are immediately visible.

Figure 1: Development of research variables per participant over time

Distribution per variable

Boxplots provide an overview of the distribution of each variable across all participants and weeks. Points outside the whiskers (1.5 × IQR) are considered outliers. Individual data points are shown semi-transparently to convey the density of the data.

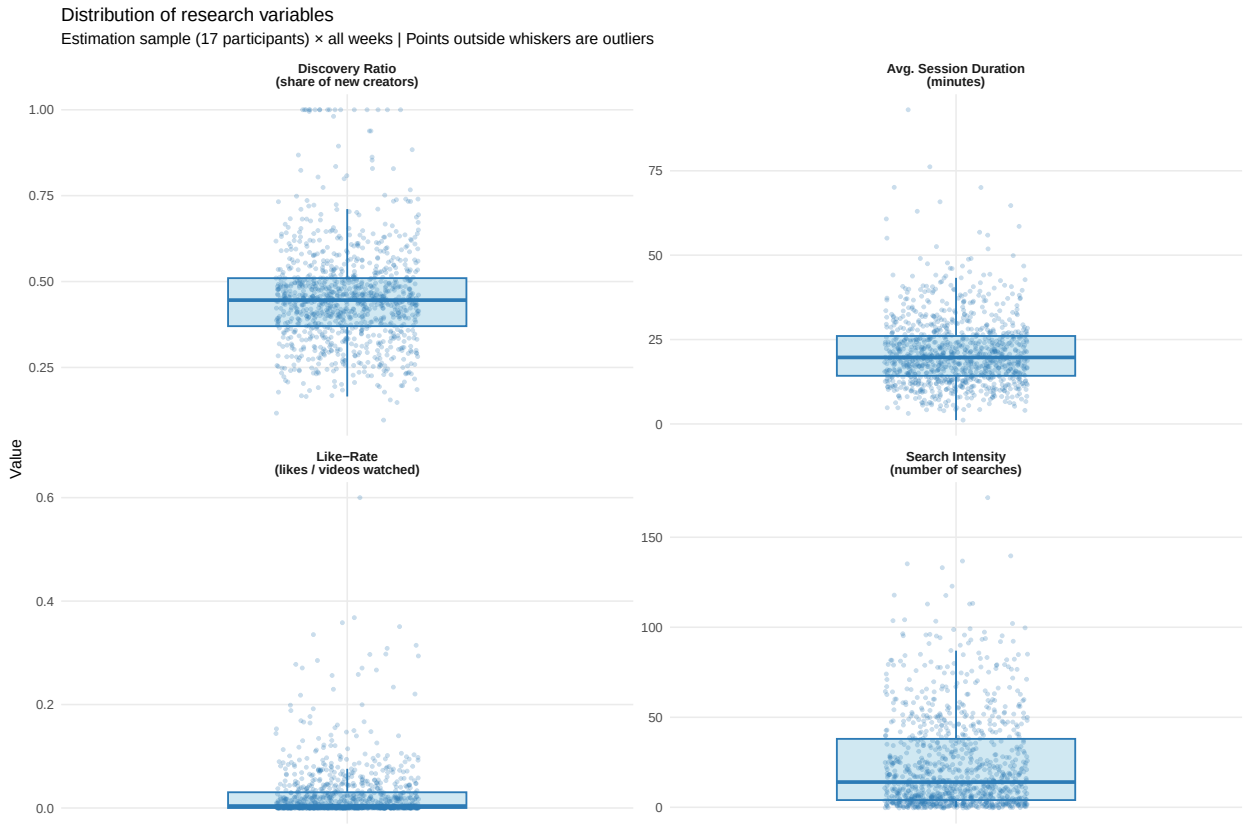

Figure 2: Distribution of research variables (all participants × all weeks)

Outlier analysis

Participant-week combinations above the 99th percentile per variable. These are values that warrant closer inspection to determine whether they are substantively correct or indicative of data issues.

Table 3: Outliers above the 99th percentile per variable

Variable	Participant	Week	Value	P99 threshold
Avg. Session Duration (min.)	p06	2026-03-02	93.077	55.677
Avg. Session Duration (min.)	p06	2025-06-30	76.182	55.677
Avg. Session Duration (min.)	p06	2025-02-03	70.102	55.677
Avg. Session Duration (min.)	p06	2024-12-09	70.055	55.677
Avg. Session Duration (min.)	p06	2025-06-16	65.807	55.677
Avg. Session Duration (min.)	p06	2026-01-05	64.693	55.677
Avg. Session Duration (min.)	p06	2025-09-15	63.002	55.677
Avg. Session Duration (min.)	p01	2025-03-17	60.738	55.677

Variable	Participant	Week	Value	P99 threshold
Avg. Session Duration (min.)	p01	2025-02-03	58.536	55.677
Avg. Session Duration (min.)	p04	2025-11-10	56.792	55.677
Avg. Session Duration (min.)	p06	2025-12-15	55.934	55.677
Like-Rate	p03	2026-02-23	0.600	0.283
Like-Rate	p16	2026-03-23	0.368	0.283
Like-Rate	p16	2026-02-09	0.358	0.283
Like-Rate	p16	2026-01-12	0.351	0.283
Like-Rate	p16	2026-02-02	0.335	0.283
Like-Rate	p03	2026-02-16	0.315	0.283
Like-Rate	p16	2026-03-09	0.309	0.283
Like-Rate	p16	2026-03-16	0.298	0.283
Like-Rate	p16	2026-02-23	0.297	0.283
Like-Rate	p16	2026-01-05	0.294	0.283
Like-Rate	p16	2025-12-15	0.285	0.283
Search Intensity	p06	2025-12-15	172.000	110.390
Search Intensity	p06	2026-01-05	140.000	110.390
Search Intensity	p06	2026-02-09	137.000	110.390
Search Intensity	p06	2025-11-24	135.000	110.390
Search Intensity	p16	2025-09-15	133.000	110.390
Search Intensity	p06	2026-02-02	123.000	110.390
Search Intensity	p06	2025-02-03	118.000	110.390
Search Intensity	p06	2025-12-01	118.000	110.390
Search Intensity	p01	2025-06-30	113.000	110.390
Search Intensity	p06	2025-06-23	113.000	110.390
Search Intensity	p06	2026-02-23	113.000	110.390

Notes per variable

Avg. Session Duration — Values up to ~93 minutes (p06) are substantively plausible. Long sessions occur when a participant uses TikTok for an extended period. There is no reason to remove these values.

Like-Rate — Values above 1.0 are mathematically impossible given the definition (likes / videos watched), but can arise due to a timing asynchrony: a video is watched in week A but only liked in week B (e.g. via a creator's profile page). Values are capped at 1.0.

Search Intensity — p06 shows consistently high search intensity (up to 172 per week), with an increasing trend towards the end of the observation period. This is substantively plausible (active search behavior) but deviates strongly from the other participants. When modeling, it may be worth considering a log-transformation of this variable due to its strong right skew.